

DIGITAL PRODUCT DESIGN

Lecture 2: Ideation and Hypothesis Development

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Outline

- Theory of ideation + opportunity development
- Ideation & refinement process
- Project matching + ideation
 - Goal is to have a team, product / problem, and ideas to improve

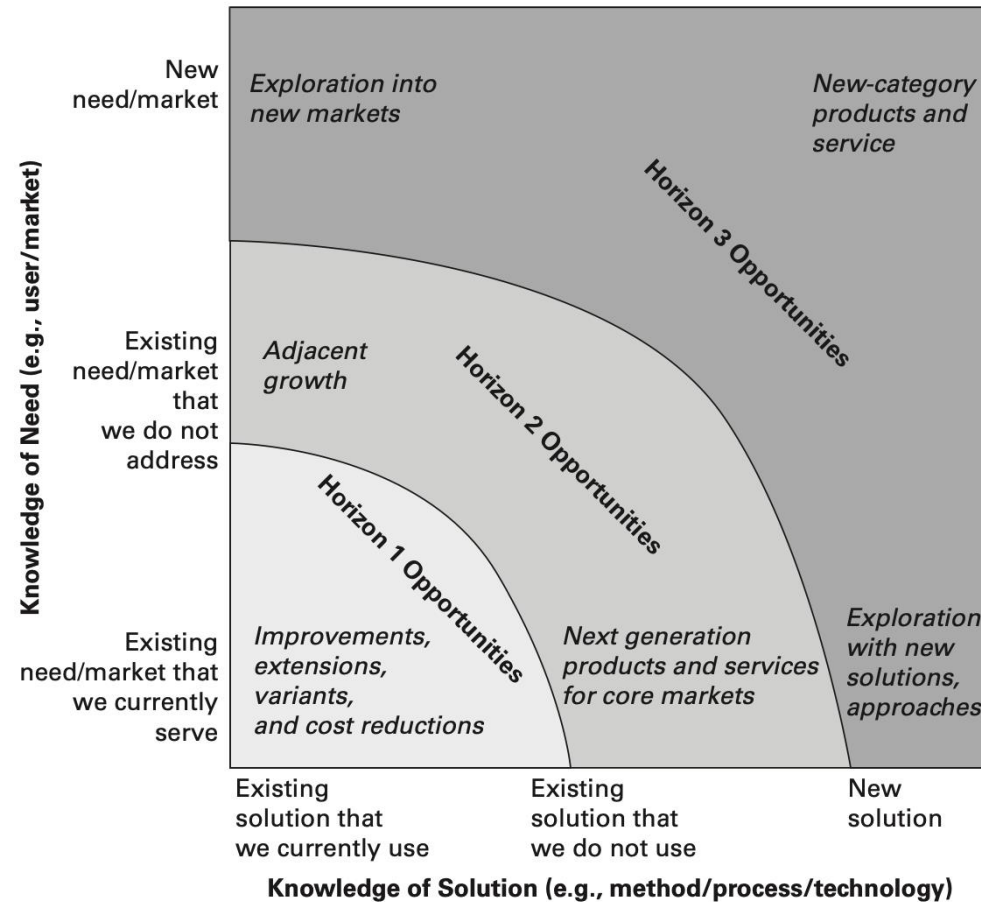
OPPORTUNITY IDENTIFICATION

* Ulrich & Eppinger Chapter 3

What is an Opportunity?

- An **opportunity** is an idea for a new product
 - A rough match for a customer need and a possible solution
 - A hypothesis about how value may be created
- A simple taxonomy of opportunities
 - How familiar is your team with the technology?
 - How familiar is your team with the market & customers?

Types of Opportunities



Opportunity Portfolio

- Opportunities are uncertain
 - What types of opportunities are more / less uncertain?
- Useful to develop several ideas in parallel
- Initial goal is to generate many opportunities and winnow down as uncertainty is resolved throughout the process

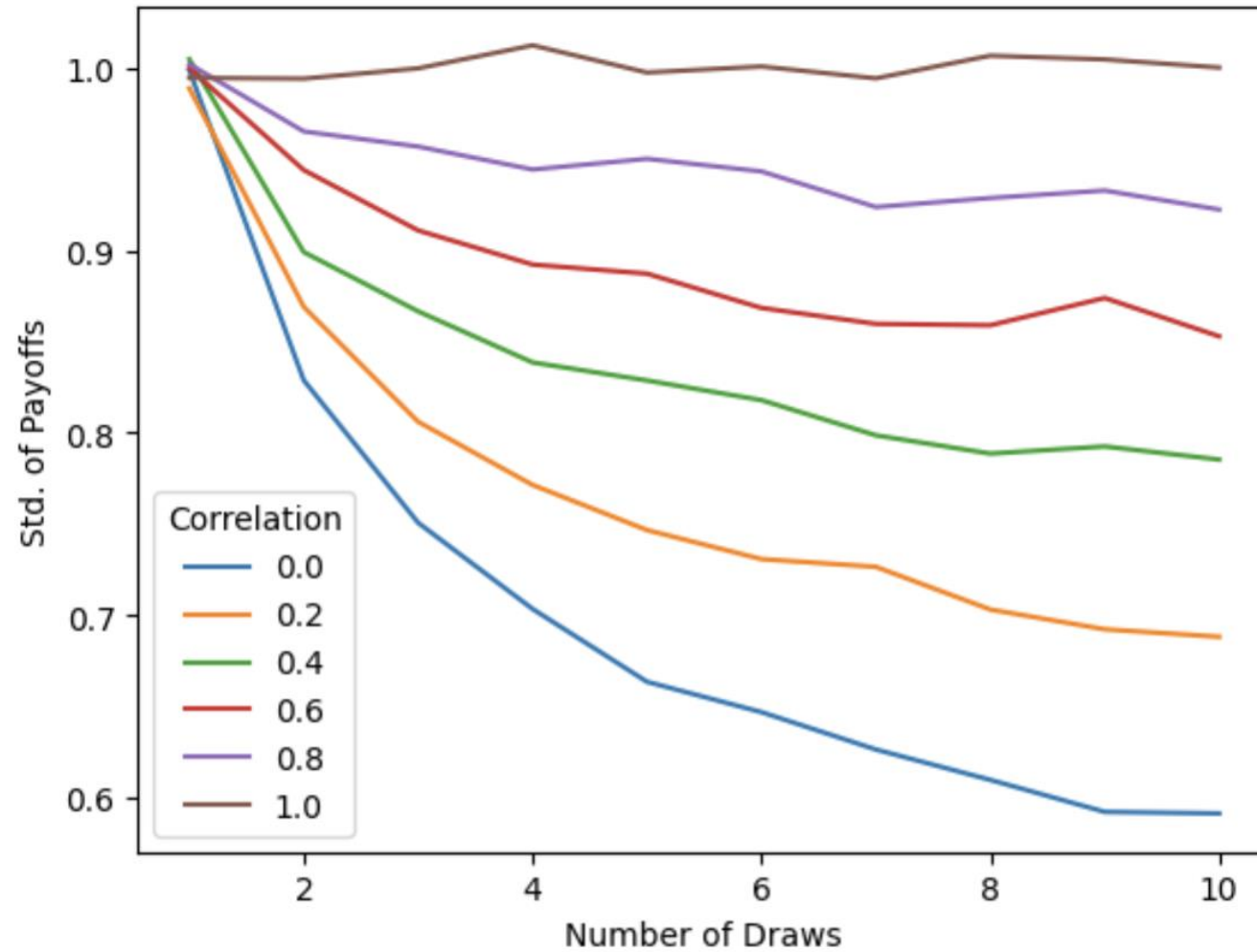
Implications of Uncertainty in Payoffs

- Generate many ideas
 - Increases the likelihood of finding an exceptional idea
- Increase variance in ideas
 - 1000 ideas that tweak the color of the website are highly correlated
 - Payoff of choosing among many uncorrelated ideas higher than payoff of choosing among many correlated ideas

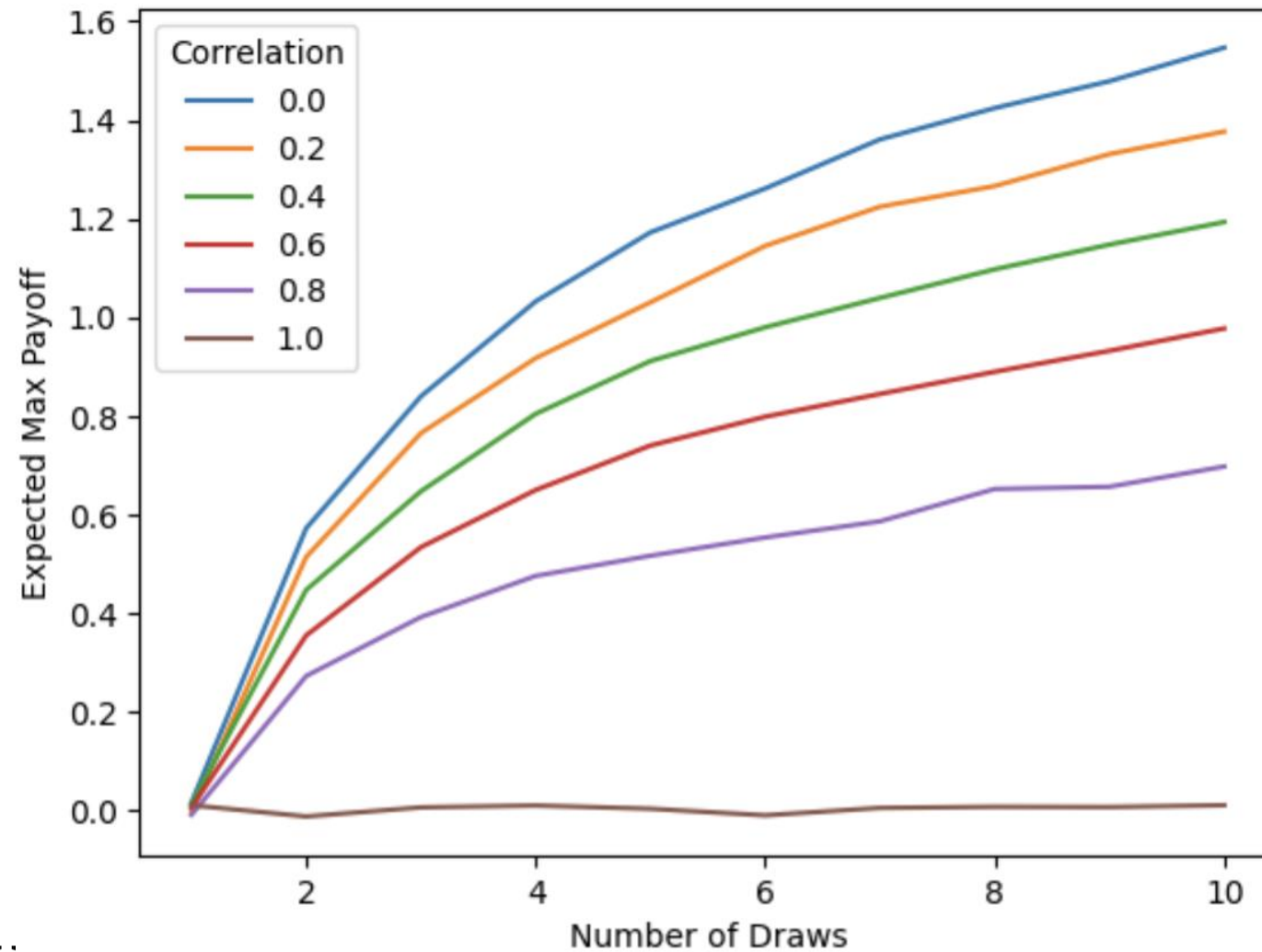
Proof-by-Simulation

- Sample from N -dimension multivariate normal distribution
 - N : Number of ideas
 - γ : Correlation between the draws
- Calculate maximum of the vector (payoff of best idea)
- Repeat this 10,000 times for various values of N and γ

Prefer Uncorrelated Ideas: Traditional Diversification



Prefer Uncorrelated Ideas: Order Statistics



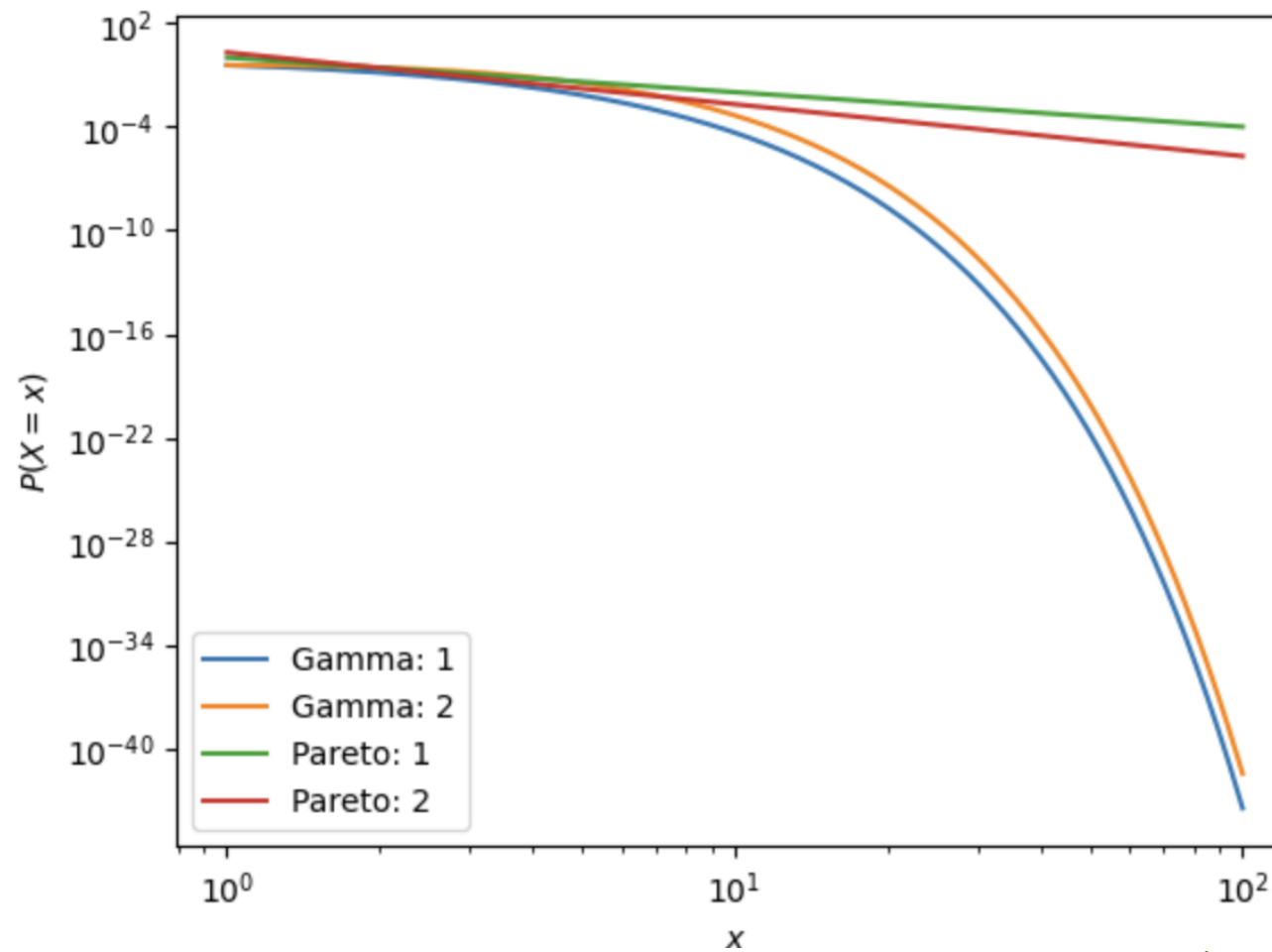
Fat-Tailed Distributions

- Thin tailed distributions: The average value provides a useful measure of the central tendency
 - Examples: Height, speed of cars, temperature
 - “Large” deviations from the mean are rare
- Fat tailed distributions: Outlier events are likely and the mean is less useful as a measure of central tendency
 - Example: Average US city population in 2000 was ~8,000
 - In digital markets, many examples of fat-tailed distributions

Power Laws and Fat Tails

- Power law describes a distribution where: $P(X = x) \propto x^{-\alpha}$
- Exponential describes a distribution where $P(X = x) \propto e^{-\alpha x}$
- $\log \frac{x^{-\alpha}}{e^{-\beta x}} = -\alpha \log x + \beta x$
 - Power law has more mass in the tails
- Power laws
 - Implies $\log P(X = x) = C - \alpha * \log x$
 - Motivates the use of log-log plots to identify power laws

Power Laws and Fat Tails, Continued



Fat-Tails in Venture Capital

How to Win in Venture Capital: Focus on the Fat Tails



The Quantified VC · Follow

16 min read · Aug 23, 2018



3.7K



16



The greatest shortcoming of the human race is our inability to understand the exponential function.

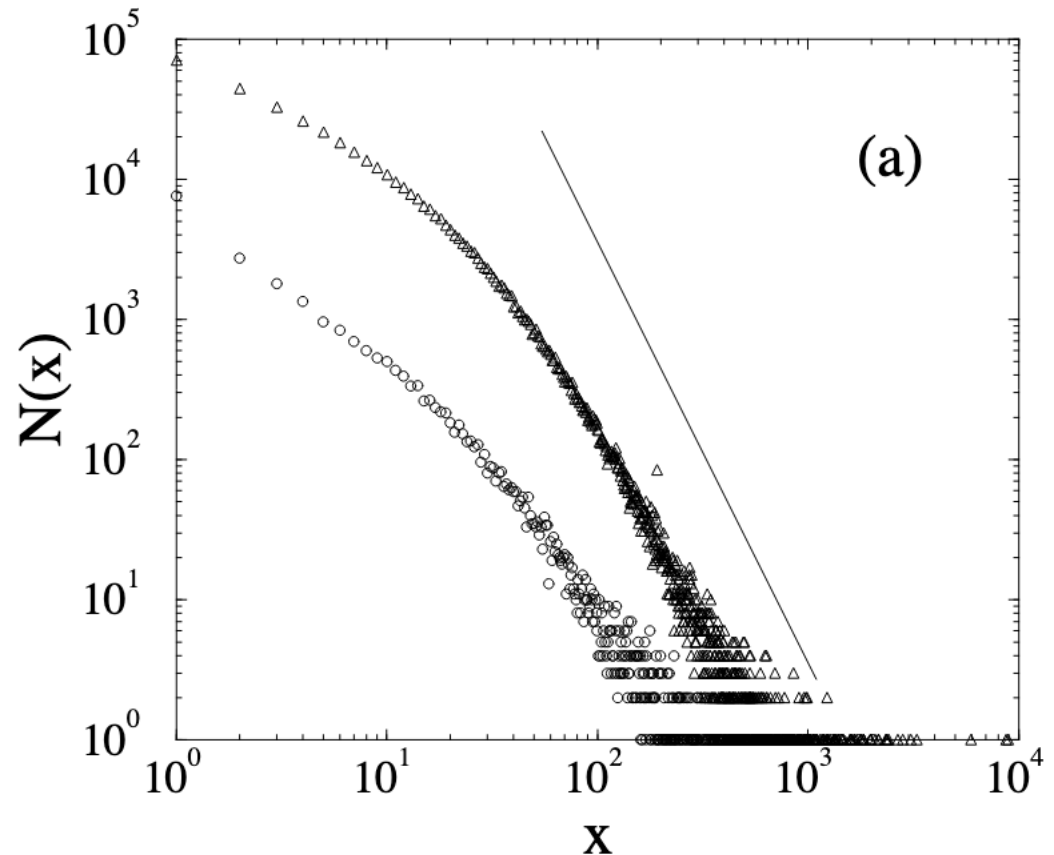
— Albert Allen Bartlett

The biggest secret in venture capital is that the best investment in a successful fund equals or outperforms the entire rest of the fund combined.

— Peter Thiel

Fat-Tails in Academic Citations

- $N(x)$ = Number of papers with x citations
- Solid line: $N(x) \propto x^{-3}$
- Tail of distributions follows power law
 - Implies fat-tails in citations



Source: Redner (1998)

Fat-Tails in Patent Citations

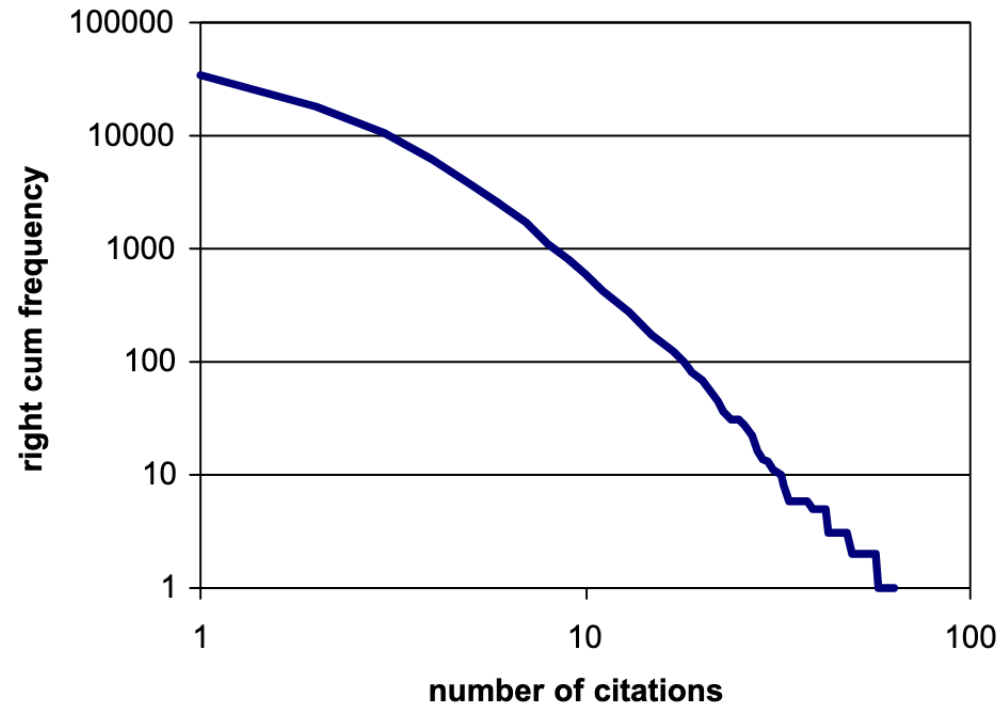
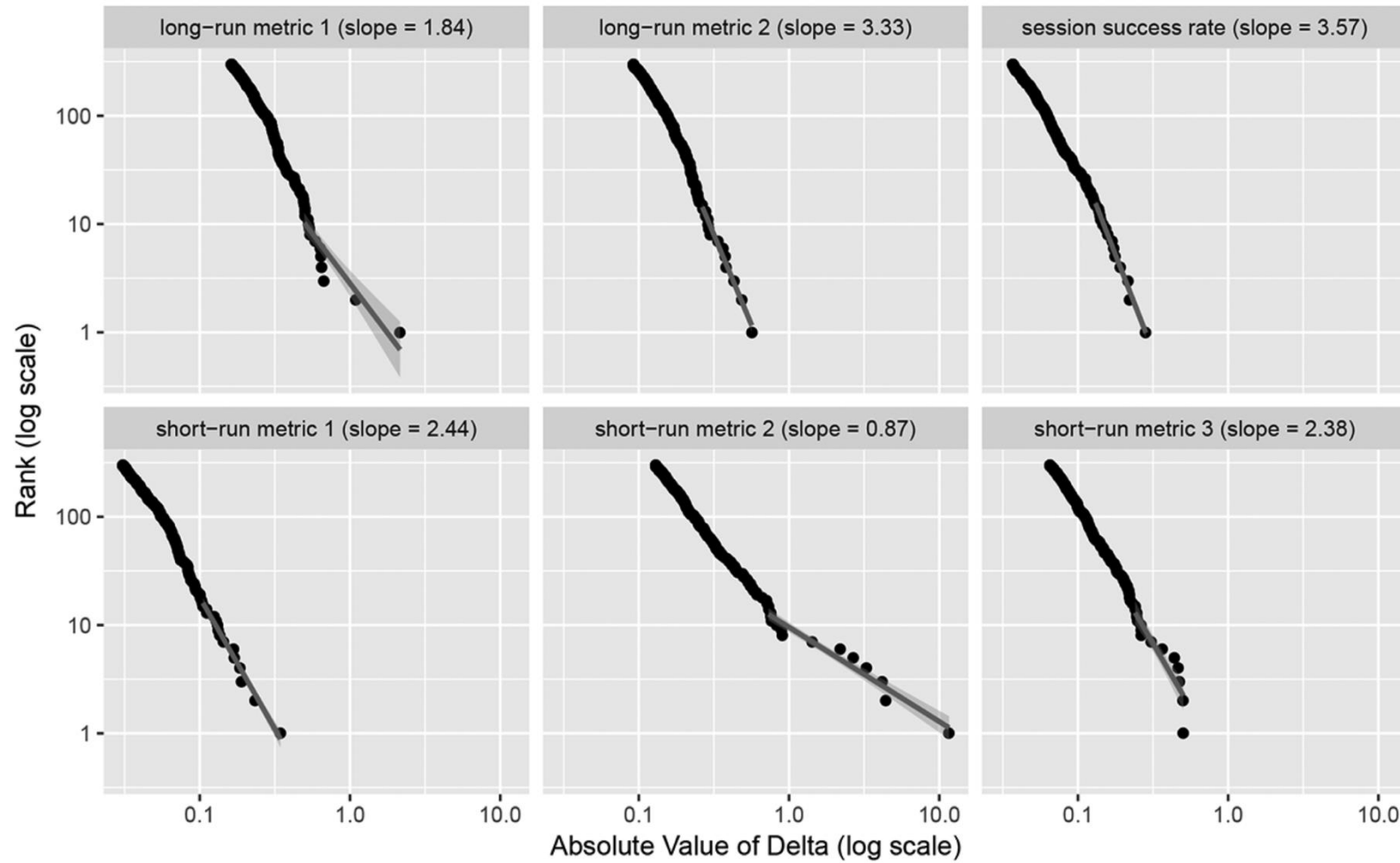


Fig. 2. Pareto plot of EPO 1989 patent citation data.

Source: Silverberg & Verspagen (2007)

Fat-Tails in Bing Experiments



Takeaways

- When your ideas have fat-tailed payoffs, there is a ton of value in identifying the outlier ideas
- We prefer uncorrelated ideas to reduce variance of payoff and increase average payoff
- Takeaway: Generate many ideas in the ideation process

OPPORTUNITY IDENTIFICATION PROCESS

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Overview of Opportunity Identification Process

- Theoretical understanding of this process is limited
 - Often more art than science
- Main understanding is:
 - Payoffs are uncertain and fat tailed
 - Implication is to generate and evaluate as many opportunities as possible

Step 1 – Set Boundaries

- Helpful to set boundaries for what we're trying to accomplish
- Boundaries should be broad
 - Type of good (physical / digital good)
 - Market / problem we to address
 - E.g. professional networking
 - Timeline
 - E.g. Launch within 12 months
- A tension between generating many uncorrelated ideas and having a more focused search for opportunities
 - Often good to err on side of too broad rather than too narrow

Step 2 – Generate Ideas

- Draw on personal experiences / passions
- Make a list of pain points you encounter in your life
- Study customers and identify latent needs
 - Interview target customer segment (both customers & non-customers)
 - Dig through bug reports / feature requests
- Imitate
 - Tweak existing innovations, push local innovations global, imitate across industries

Step 3 – Screen Opportunities

- Solicit feedback internally and externally
 - Vote internally, ask your friends + family
- Goal here is to remove opportunities that are not promising
 - Remove “losers” rather than identify the “winner”
- Don’t remove ideas that are polarizing
 - Strong opinions often suggest a promising opportunity
- Be thoughtful with how you aggregate opinions
 - Anonymous / online voting can encourage participation and expression of diverse viewpoints

Step 4 – Develop Opportunities

- Market research & trends
 - How big is the market? Is it growing / shrinking? What is the competitive landscape?
- Interview potential customers
 - Is there excitement about the opportunity?
 - Formal survey techniques (e.g. conjoint analysis)
- Analyze feasibility
 - Estimate cost (in dollars and time) of development / production
 - Is our innovation patentable / proprietary / easily replicable?

Step 5 – Select Exceptional Opportunities

- We now have the information needed to identify best opportunities
- Many, many, many frameworks that can be applied
- One example: Real-Win-Worth it
 - Is the opportunity real? Does the market exist and will our opportunity address the need in the market?
 - Can you win? Can you create a sustainable competitive advantage through patents / technology / network effects / know-how?
 - Is the opportunity worth it financially?

Step 6 – Reflect

- Did we do a good job in generating and selecting opportunities?
- Do we feel good about the outcomes?
- If we're not excited about the opportunity, who will be?

PROJECT MATCHING

Project Matching + Ideation

- If you have a product you want to use for project, stand at perimeter of class or raise your zoom hand
 - If online, introduce your product and note the breakout room
- Spend 5 minutes talking with each other, forming teams of 4
- Work through the opportunity identification process
 - Set boundaries for the process
 - Spend 10-15 minutes generating ideas
 - Cull the ideas to most promising
 - Develop the ideas (or think about how you would develop)

Share your Ideation Experience

- What worked?
- How many ideas did you generate?
 - Are they correlated?
- What uncertainty remains?
- Do you have a project that you are excited to work on?

QUESTIONS